Syntecxhub AI Internship — Week 1, Project 2

Rule-Based Expert System: Symptom Checker

A simple expert system that mimics human reasoning using IF-THEN rules and forward chaining, with a full explanation (log) of how it reached its conclusion.

Files

- `expert_system.py` — the complete program

How it works

1. **Rules**: a small knowledge base of `IF [symptoms] THEN [condition]` rules.
2. **Facts**: the symptoms you confirm having (via yes/no prompts).
3. **Forward chaining**: the engine repeatedly checks all rules against known facts. Any rule whose conditions are fully met "fires," adding its conclusion as a new fact — which can then trigger further rules (multi-step/chained inference).
4. **Reasoning log**: every fired rule is recorded and shown to the user, so the system's logic is fully transparent.

How to run

```
python3 expert_system.py
```

Answer each symptom prompt with `y` (yes) or `n` (no). At the end you'll see the reasoning steps and the final conclusion(s).

Example rules included

If symptoms include...	Then concludes...
cough, sore throat	Common Cold
fever, body ache, fatigue	Flu

fever, cough, shortness of breath	Respiratory Infection
sneezing, itchy eyes, runny nose	Allergy
Common Cold + fever (chained)	See a doctor
Flu + shortness of breath (chained)	Seek urgent care

Notes

This is an educational demo for an AI internship task and is **not** real medical advice.

Requirements

- Python 3.x (no external libraries needed)